

Name: _____

Homework 7 | Math 253 | Huchala

Due Wednesday of Week 8 at the start of class

Complete the following problems and submit them in class. 8 points are awarded for thoroughly attempting every problem, and I'll select four problems to grade on correctness. Enough work should be shown that there is no question about the mathematical process used to obtain your answers.

In problems 1–3, find a power series for the function using power series multiplication and determine its interval of convergence.

1. $\left(\frac{1}{1-x}\right)\left(\frac{1}{1+x}\right).$

2. $\left(\frac{1}{1-x}\right)\left(\frac{1}{(1+x)^2}\right).$

3. $\left(\frac{1}{(1-x)^2}\right)\left(\frac{1}{(1+x)^2}\right).$

In problems 4–5, find a power series for the given function using calculus and determine its interval of convergence.

4. $\ln(1+x^2).$

5. $\frac{1}{(1+2x)^4}.$

6. Let $f(x) = \sum_{n=0}^{\infty} c_n x^n$ on $(-R, R)$, so that the series does not converge at $x = R$. Is it possible for the series for $f'(x)$ to converge at $x = R$? Give an example of a series where this is true or justify why no such series can exist.

7. Let $f(x) = \sum_{n=0}^{\infty} c_n x^n$ on $(-R, R)$, so that the series does not converge at $x = R$. Is it possible for the series for $\int f(x) dx$ to converge at $x = R$? Give an example of a series where this is true or justify why no such series can exist.

8. Let $f(x) = \sin(x)$.

a) What are $f(0)$ and $f'(0)$?

b) Suppose $f(x) = \sum_{n=0}^{\infty} c_n x^n$. What are c_0 and c_1 ?

c) Since $f''(x) = -\sin(x) = -f(x)$, the coefficient sequence (c_n) satisfies a recurrence. Write that recurrence down.

d) Find an explicit formula for c_n . (Hint: handle even and odd n separately).

e) What is the interval of convergence for $\sum_{n=0}^{\infty} c_n x^n$?